

All Rise

AI Courtroom Simulation

Practice legal reasoning against an AI prosecutor that never forgets what you said.

Live Demo	courtroom-ai-eight.vercel.app
GitHub	github.com/sriksven/courtroom-ai
Builder	Krishna Venkatesh
Role	Solo, Full Stack, AI Architecture, Design

Inspiration

Most people will face a moment where they need to advocate for themselves, whether in a disciplinary hearing, a workplace dispute, or a high-stakes conversation. But legal reasoning and structured argumentation are never formally taught. You either pick them up through experience or you do not.

Access to mock trial practice is expensive and exclusive. Law students pay for coaching. Debate teams require membership. The goal was to build something that made argumentation training free, accessible, and genuinely engaging for anyone, anywhere.

The core question: can an AI system be designed not to help you, but to challenge you? Not an assistant but an adversary. One that remembers what you said, exploits your weaknesses, and forces you to think harder every round. That question became All Rise.

Problem Statement

Legal reasoning and argumentation are critical life skills that most people never get to practice in a safe, structured environment. Mock trial programs, debate coaching, and legal literacy education are largely inaccessible to students without institutional support or financial resources.

At the same time, most AI tools are designed to assist users, not challenge them. There are very few systems that put a user under genuine intellectual pressure, force them to construct and defend a position, and evaluate the quality of their reasoning objectively.

The result is a gap: millions of people lack the ability to argue effectively under pressure, and no accessible tool exists to help them build that skill.

What It Does

All Rise is a fully realized adversarial AI courtroom. You play the defendant. The charge is absurd and formally stated. You argue your innocence across structured trial phases against three independent AI agents running simultaneously on the same trial state.

Agent	Model	Role
Reginald P. Harrington III	Groq llama-3.3-70b	Prosecutor with ReAct loop and per-trial memory
Judge Constance Virtue	OpenAI GPT-4o	Silent judge who scores 4 dimensions at verdict
The Strategist	Groq llama-3.3-70b	Defense assistant, adaptive hints, never repeats

You play the fourth role: the defendant. The trial takes roughly 4 minutes. There is no guaranteed outcome. You can lose.

Key features include:

- AI charge generator: type any topic and the AI drafts a formal dramatic charge
- Three interaction modes: text, auto-voice (browser STT/TTS), and full LiveKit WebRTC voice
- Difficulty selector: Easy, Normal, and Hard modes affecting prosecutor behavior
- Prosecutor mood escalation: confident in round 1, rattled by round 4, unhinged by round 8
- Court intervention at round 8: judge breaks silence mid-trial to order wrap-up
- Hint counter: 3 hints per trial from The Strategist, disabled in closing phase
- Animated verdict page with staggered score reveal and fallacy breakdown
- Judge scores Argument Strength, Evidence, Logic, Persuasion (max 40 pts, 28 to win)

Architecture and Technical Implementation

Frontend

React 18 with Vite and Tailwind CSS, deployed on Vercel. Trial state is managed through a useReducer state machine across 7 distinct phases. There is no React Router as page transitions are CSS-based. Every chat bubble has its own play/pause/stop TTS player powered by OpenAI tts-1-hd with three distinct voices: Reginald on onyx, the Judge on shimmer, and the defendant on alloy.

Agent Architecture

Three agents with conflicting goals run on a shared trial state. Each agent has its own memory partition inside a TrialMemory map keyed by trial ID:

```
TrialMemory prosecutorMemory (round summaries, attack strategy, evidence used, fallacy
log) judgeMemory (accumulated scores, fallacy list) defenseMemory (hint history, user
arguing style)
```

The prosecutor runs a full ReAct loop every round:

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OBSERVE -> recallWeaknesses(defenseText) detectFallacy(defenseText)
recallAttackStrategy(memory) getUnusedEvidence(memory) ACT -> groqChat(prompt + tool
results) WRITE -> recordEvidenceUsed updateAttackStrategy addFallacy -> judgeMemory
addRoundSummary
```

Infrastructure

The agent process runs on Railway connected to LiveKit Cloud over WebSocket. If Railway goes down, all API calls route silently to stateless Vercel serverless functions. The user never notices the fallback. A GitHub Actions CI pipeline runs all 79 Vitest and React Testing Library tests on every push.

Tech Stack

Layer	Technology
Frontend	React 18, Vite, Tailwind CSS
LLM (prosecutor, hints)	Groq llama-3.3-70b-versatile
LLM (judge verdict)	OpenAI GPT-4o with structured JSON output
Text-to-Speech	OpenAI tts-1-hd (3 voices)
Speech-to-Text	Web Speech API (browser-native)
Voice rooms	LiveKit WebRTC
Frontend deploy	Vercel (serverless functions)
Agent deploy	Railway
Testing	Vitest, React Testing Library, Playwright
CI	GitHub Actions

Challenges

Three agents on one trial state: The prosecutor needed to remember round 1 in round 5. The judge needed to stay silent while accumulating context. The defense assistant needed to adapt to arguing style without handing out pre-written answers. Three very different memory models on the same shared state.

Fallback architecture: The full experience runs on Railway but the app needed to work perfectly if that process was down. The stateless Vercel fallbacks had to be functionally identical from the user's perspective, just without persistent memory.

Prosecutor mood escalation: Getting a large language model to feel genuinely confident in round 1 and rattled in round 4 without it feeling mechanical took several iterations of prompt tuning across difficulty levels.

Judge scoring rubric: Calibrating the rubric so that a well-argued defense actually wins and a poorly argued one actually loses required careful iteration. GPT-4o structured JSON output made scores deterministic and auditable.

Voice mode synchronization: The auto-voice while-loop had to coordinate browser STT, API calls, and TTS playback without race conditions or overlapping audio across three distinct agent voices.

Accomplishments

- Three independent AI agents with conflicting goals producing a genuinely uncertain outcome every trial
 - A prosecutor that remembers what you said in round 1 when it attacks you in round 5
 - A judge scoring system that is objective, consistent, and auditable through structured GPT-4o JSON
 - A fallback architecture so robust the user cannot tell whether the Railway agent is live or down
 - Three fully functional interaction modes including real-time LiveKit WebRTC voice
 - 79 automated tests covering the full trial state machine and agent flow
 - A four-minute experience that teaches critical thinking without feeling like education
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What We Learned

Building adversarial agents is fundamentally different from building assistants. The goal is conflict, not helpfulness. Designing a system where agents actively work against each other while still producing a fair and coherent outcome taught more about prompt engineering, state management, and multi-agent coordination than any previous project.

Reliability engineering mattered more than expected. A multi-agent system with external API dependencies and a Railway-hosted agent process has many failure points. Designing the fallback architecture early saved the project from critical failures during the event.

The user experience depends heavily on things that have nothing to do with the AI itself. Phase-aware loading messages, per-bubble TTS controls, mood escalation, and the animated verdict reveal are what make the trial feel real. The surrounding product has to hold the experience together.

Social Impact

Legal reasoning is a skill most people never get to practice safely. All Rise makes that practice free, accessible, and genuinely engaging. The prosecutor detects logical fallacies in real time. The judge scores reasoning objectively across four dimensions. Every trial is a low-stakes environment to get better at thinking under pressure.

These skills matter far beyond a courtroom. They apply to job interviews, academic disputes, workplace conflicts, negotiation, and everyday advocacy. Students without access to debate programs or mock trial coaching can use All Rise to build the same skills others pay to develop.

The platform is free, requires no account, runs entirely in the browser, and completes a full trial in four minutes. The barrier to entry is zero.

What's Next for All Rise

- Multiplayer mode: two defendants argue the same charge simultaneously, the judge picks a winner

- Case history and progression: track scores across trials, watch your fallacy rate drop over time
- Educator dashboard: teachers assign charges, review transcripts, use scored verdicts as a classroom tool
- Structured debate mode: expand into job interview prep, negotiation simulations, academic defense practice
- Custom prosecutor personas: choose your opponent's personality and arguing style
- Mobile app: the voice pipeline is already built, packaging it natively is the logical next step